



NATIONAL INSTITUTE OF TECHNOLOGY
KURUKSHETRA-136119
B. Tech. (R&A) – 4th Semester
END SEMESTER EXAM; AUGUST – 2026
Date of Exam: 11th August 2026

RAPC-212

Time: 180 Minutes

Manufacturing Processes

Max. Marks: 50

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1 (a) Classify crystal imperfections and explain their significance. Explain the formation and significance of grain boundaries and twin boundaries in engineering materials. (5)

(b) Differentiate between up milling and down milling and explain why down milling is generally preferred in modern CNC milling machines. (5)

Q-2: Define extrusion and drawing processes. Explain their working principles, types, advantages, limitations, and applications. A strip of steel having an initial thickness of 20 mm is reduced to a final thickness of 15 mm in a single pass rolling operation. The width of the strip remains constant at 200 mm. The average flow stress of the material during rolling is 250 MPa. The roll radius is 300 mm and the rolls rotate at 100 rpm. Calculate, (a) Percentage reduction in thickness, (b) True strain developed during rolling, and (c) Rolling Force. (10)

Q-3: a) Explain the functions of the runner and riser in a casting system. Explain the complete working procedure of Investment Casting with a neat flow diagram. (5)

b) Explain the complete sequence of Powder Metallurgy processing with a suitable flow diagram. Explain the functions of binders and discuss the important characteristics of a suitable binder. (5)

Q-4: a) Classify the major Additive Manufacturing (AM) processes. Explain the basic steps involved in an AM process. Explain the working principle of material-extrusion-based Additive Manufacturing with a neat schematic diagram. (5)

b) Explain the working principles of explosive forming and hydraulic forming with suitable sketches. (5)

Q-5: Explain the working principle and detailed process mechanism of Electrical Discharge Machining (EDM) with a neat schematic diagram. Explain the different types of electrode wear encountered in EDM. What are the important functions of dielectric fluid in EDM? Discuss the desirable properties of a dielectric used in EDM. (10)

End of Question Paper

